CSE2023 - ASSIGNMENT #4

Solve the following questions from course book. (Discrete Mathematics and Discrete Mathematics and Its Applications, 7th Ed. by Kenneth Rosen)

Note that we may grade selected questions from HWs.

* Show all your works.

1) Prove that $2^n > n^2$ if n is an integer greater than 4.

2) How many positive integers between 22 and 121 inclusive
    a. are divisible by 3?
    b. are divisible by 4?

3) How many bit strings of length 14 contain at least six 1s and at least five 0s?

4) What is the minimum number of students, each of whom comes from one of the 40 states, who must be enrolled in a university to guarantee that there are at least 80 who come from the same state?

5) The Turkish alphabet contains 21 consonants and 8 vowels. How many strings of ten lowercase letters of the Turkish alphabet contain
    a. exactly three vowel?
    b. exactly four consonants?
    c. at least two vowel?
    d. at least three vowel?
    e. at least three consonants?

Submission Instruction (10p)
Please zip and submit all your files using filename YourNumberHW2.zip
(ex: 150629573HW2.zip) to Canvas system (under Assignments tab).

Your zip file should contain the following:
    1. Single PDF file for solutions (150629573HW2.pdf)

Notes:

1. Write your name and student ID on each sheet.
2. No late submission will be accepted